
Notions

- For a set E , the cardinal and cardinal of collection of subsets are finite, denoted by

$$|E| \quad 2^{|E|}$$

- Given a collection of sets $\mathcal{A}$,

$$\min(\mathcal{A}) \quad \max(\mathcal{A})$$

are well-defined.

Chapter 1 Definitions

1.1 Independent Set and Circuits

1.1.1 Matroid

Definition 1.1.1.1 (Whitney,1935) A matroid M is an ordered pair $(E, \mathcal{I})$ consisting of a finite set E and a collection $\mathcal{I}$ of subsets of E having the following three properties:

- $\emptyset \in \mathcal{I}$;
- (Hereditary) If $I \in \mathcal{I}$, $I' \subseteq I$, then $I' \in \mathcal{I}$;
- (Independence Augmentation) If I_1 and I_2 are in $\mathcal{I}$ and $|I_1| < |I_2|$ (which implies $I_2 - I_1 \neq \emptyset$), then there is an element e of $I_2 - I_1$ such that

$$I_1 \cup \{e\} \in \mathcal{I}$$

The members of $\mathcal{I}$ are called **independent sets** of M . E is called the **ground set**.
If $I \notin \mathcal{I}$, I is dependent.

♠ **Proposition 1.1.1.2** Let E be the set of column labels of $m \times n$ matrix over F . And let $\mathcal{I}$ be the set X of subsets of E , whose vectors indexed by the X is linear independent in $V(m, F)$, then

$$(E, \mathcal{I})$$

is a matroid.

◦ **Proof** ...

□

1.1.2 Circuits

Definition 1.1.2.1 ...

1.2 Bases

◇ **Lemma 1.2.0.1** If B_1, B_2 are bases of Matroid M . Then

$$|B_1| = |B_2|$$

◦ **Proof** ...

□

◇ **Lemma 1.2.0.2** ...

◦ **Proof** ...

□

► *Theorem* ...

◦ **Proof** ... □

◇ **Lemma 1.2.0.3** All the members of $\mathcal{B}$ have the same cardinality.

◦ **Proof** ... □

1.3 Rank

1.4 Closure

1.5 Geometric Representation of Matroid of Small Rank

Chapter 2 Matroid Intersection

2.1 Matroid Intersection Theorem

► **Theorem** (Matroid Intersection Theorem, Edmond) Let

$$M_1 = (S, I_1) \quad M_2 = (S, I_2)$$

be matroids, with rank functions r_1, r_2 . Then the maximum size of set in $\mathcal{I}_1 \cap \mathcal{I}_2$ is equal to

$$\min_{U \subseteq S} (r_1(U) + r_2(S - U))$$

◦ **Proof** ...

□

Chapter 3 Duality

Chapter 4 Minors

Chapter 5 Connectivity

Chapter 6 Representable Matroids

Chapter 7 Constructions

Chapter 8 Higer Connectivity

Chapter 9 Binary Matroid